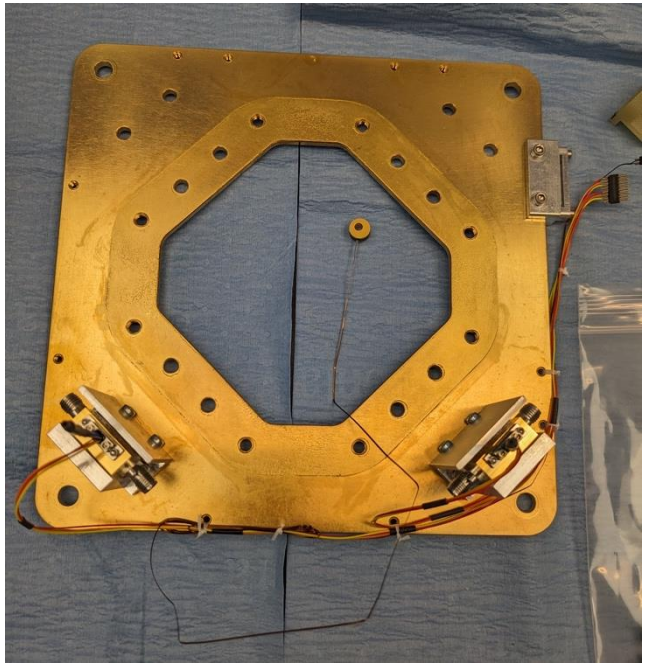
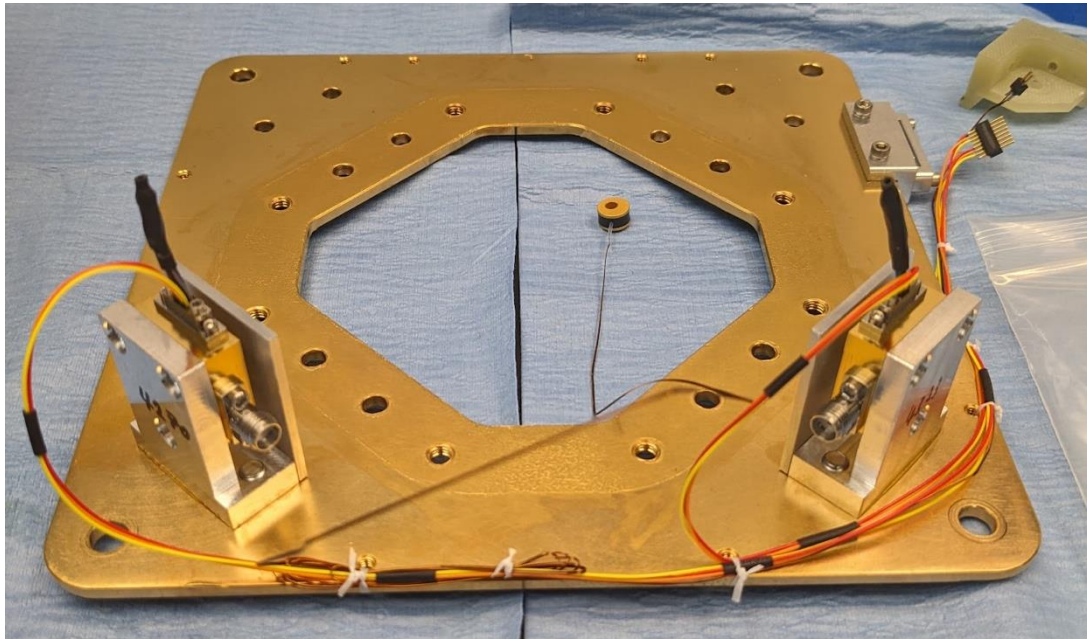
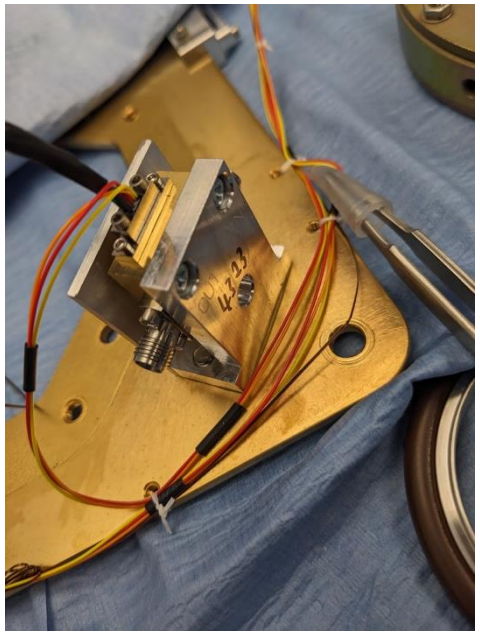
LNA cable wire harnesses

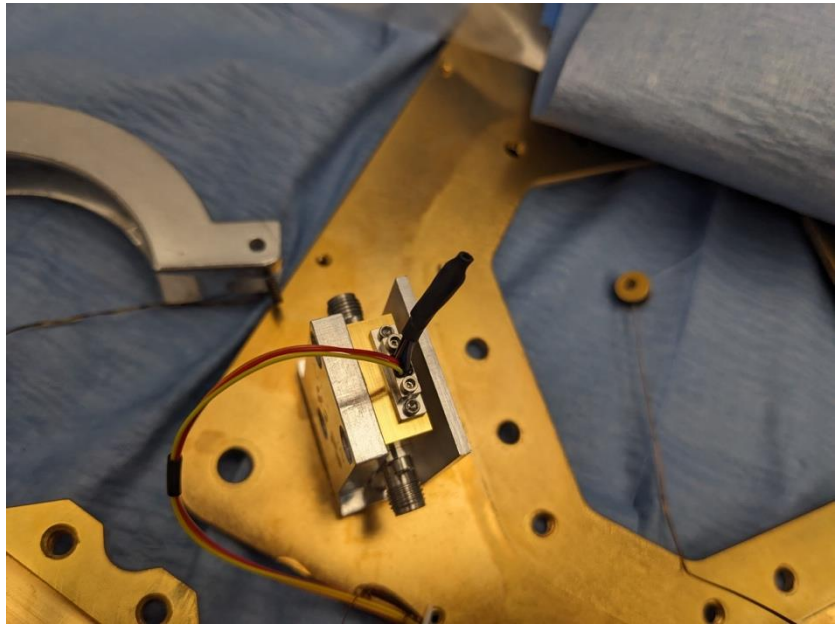
Purpose

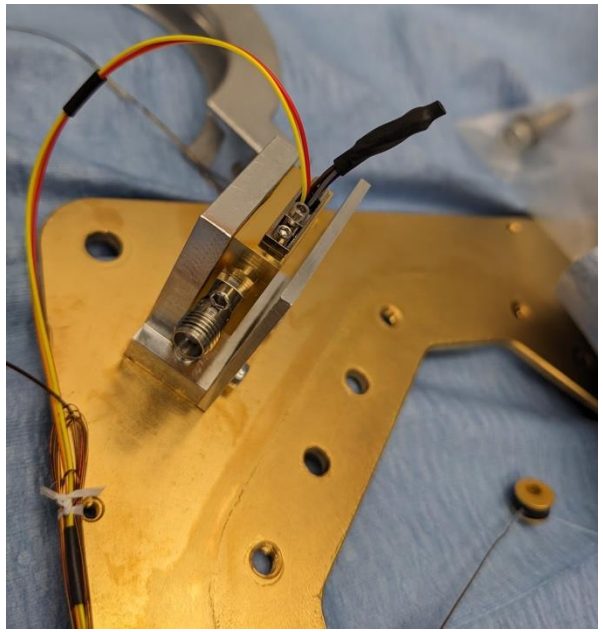
Power and control wires to the Low Noise Amplifiers (LNAs) inside the feed.



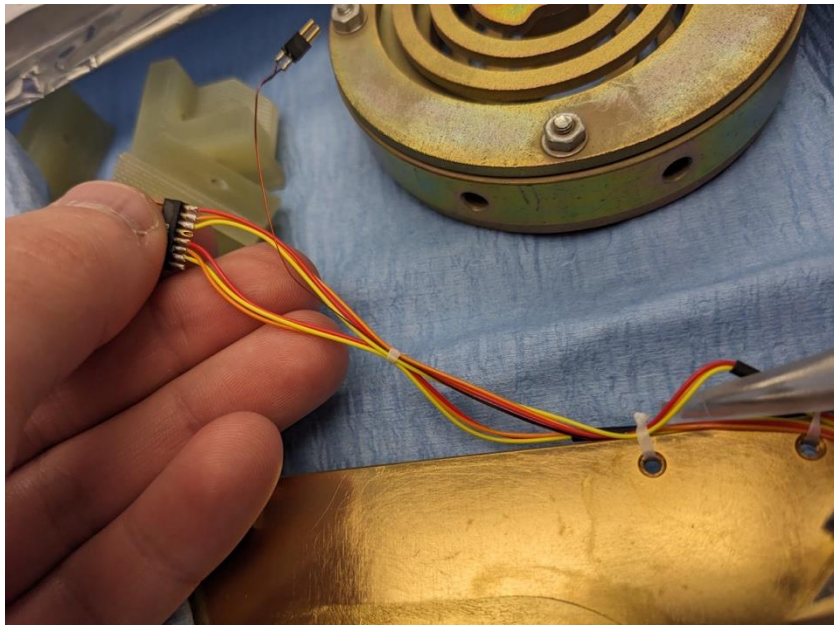


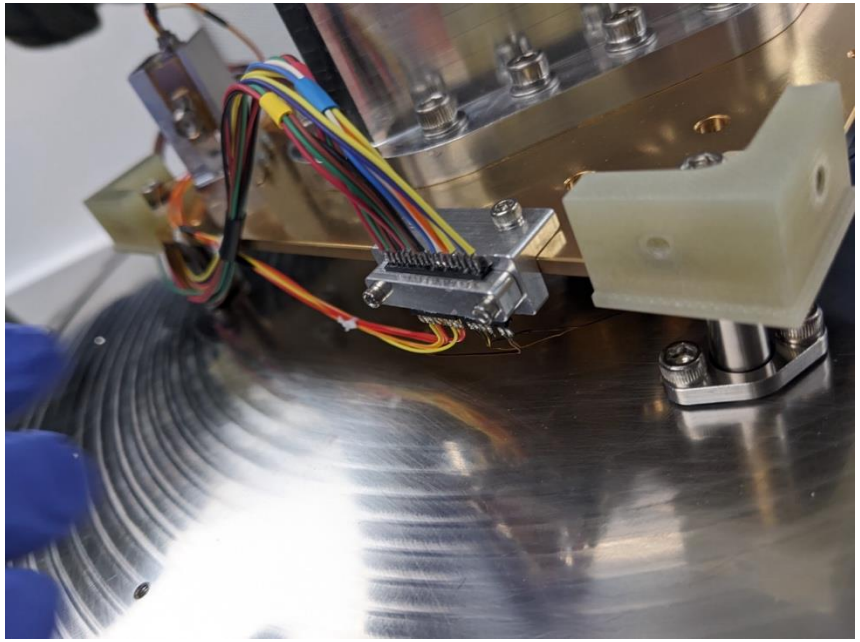


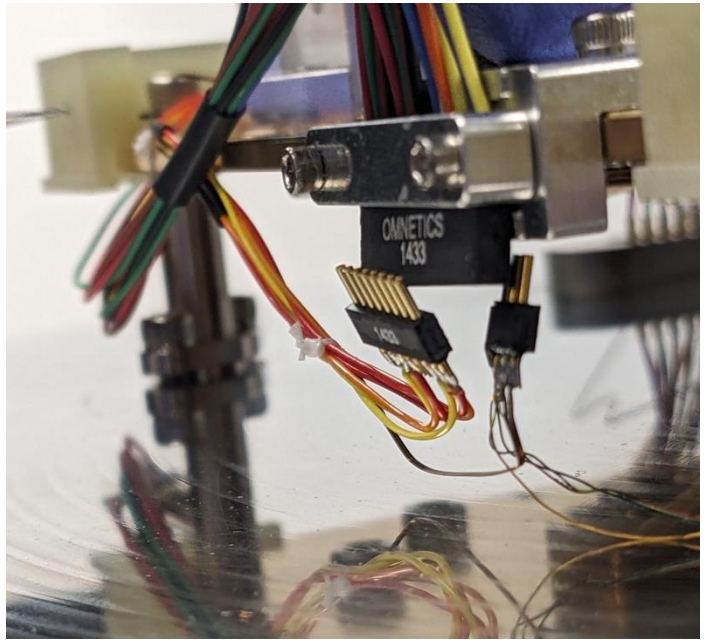




Detail images of LNA cables installed in a feed



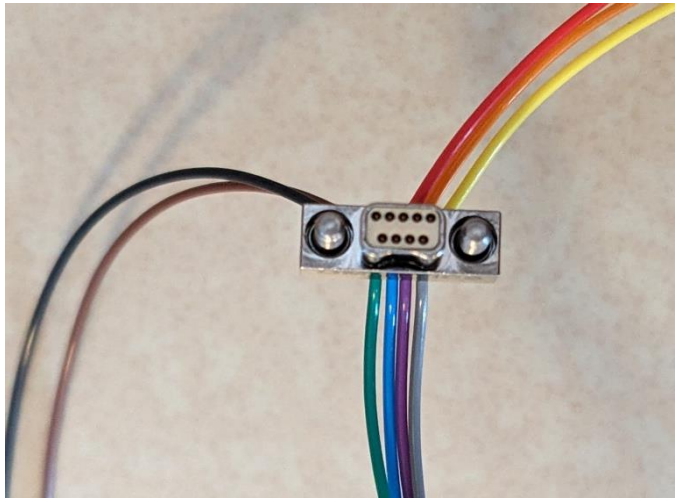




Outline of steps

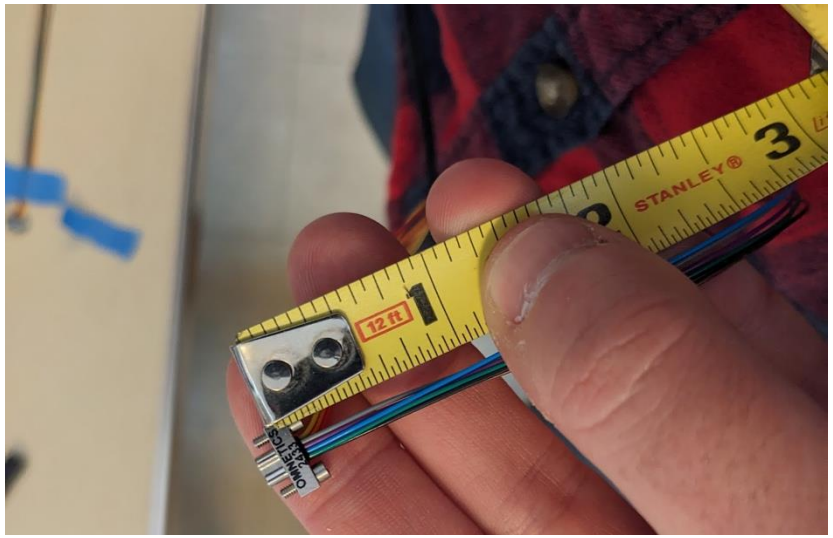
1. Trim and seal excess wires
2. Cut to length (13.5” for the y-pol cable, 9.5” for the x-pol cable)
3. Add heatshrink wrap every 2-3” along bundle
4. Solder x-pol cable and y-pol cable to 8-pin connector

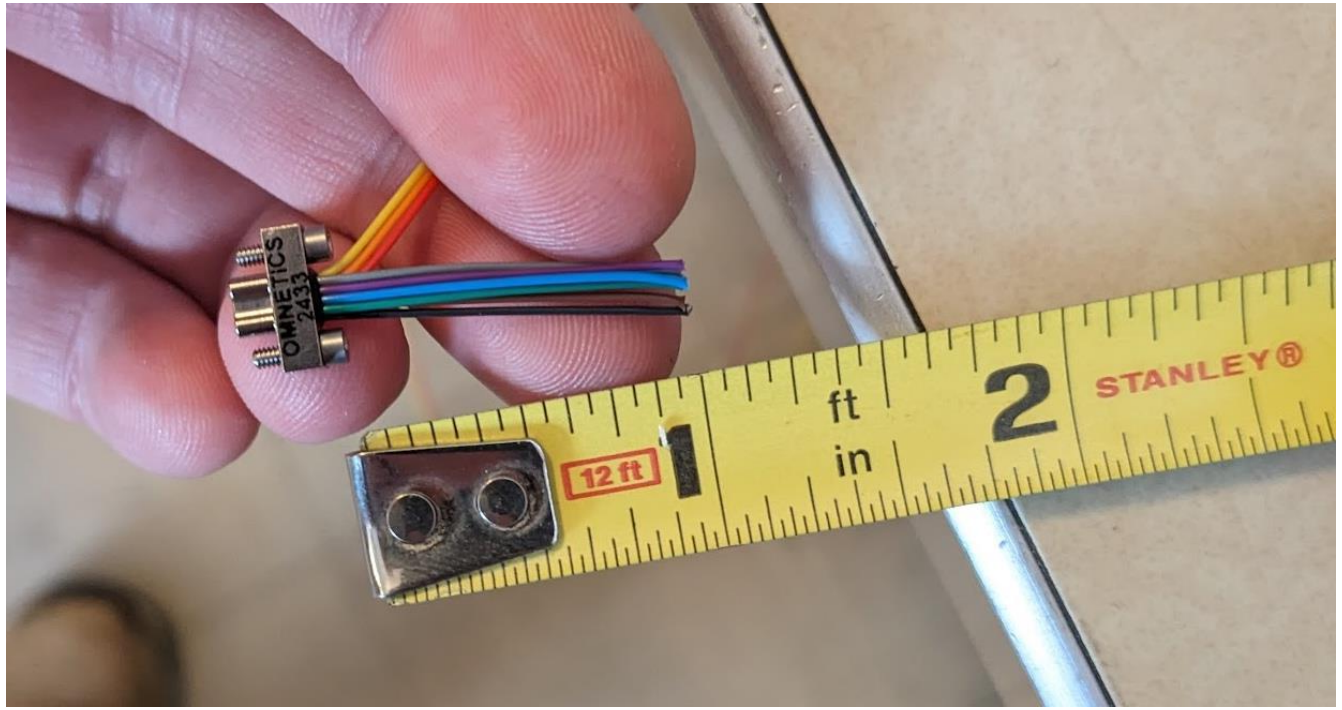
The off-the-shelf cable includes 9 wires of different colors. For our purposes, we only need 3 wires.
Separate the black, brown, green, blue, purple, and gray (excess) wires from the red, orange, and yellow wires.



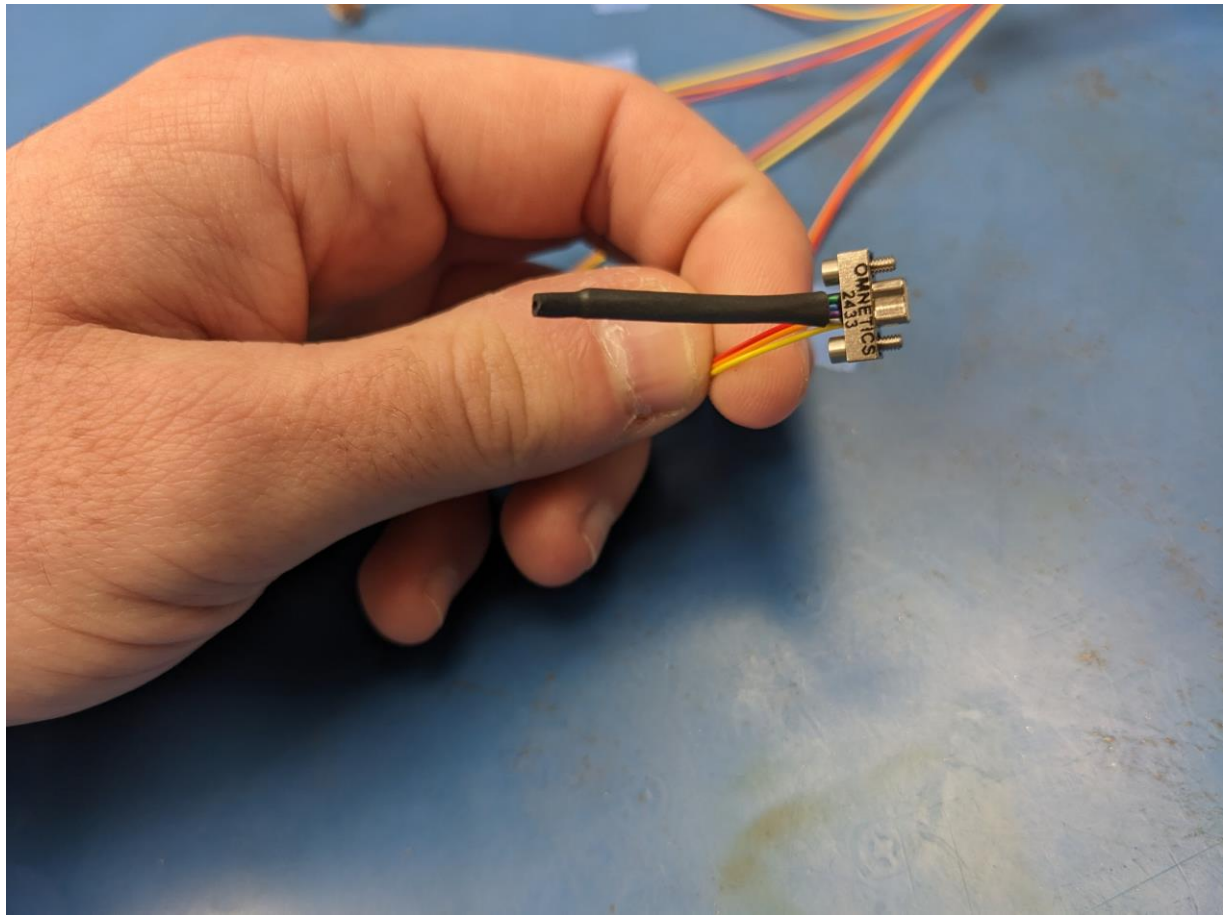
Trim the excess wires so that about 1 inch, measured from the base of the connector, remains. This does not need to be exact.

Leave the red, orange, and yellow wires long for now.





Cut about an inch of 3/32” shrink wrap tube.
Completely cover the bundle of trimmed excess wire with the heatshrink tube.
Apply the heat gun to seal.



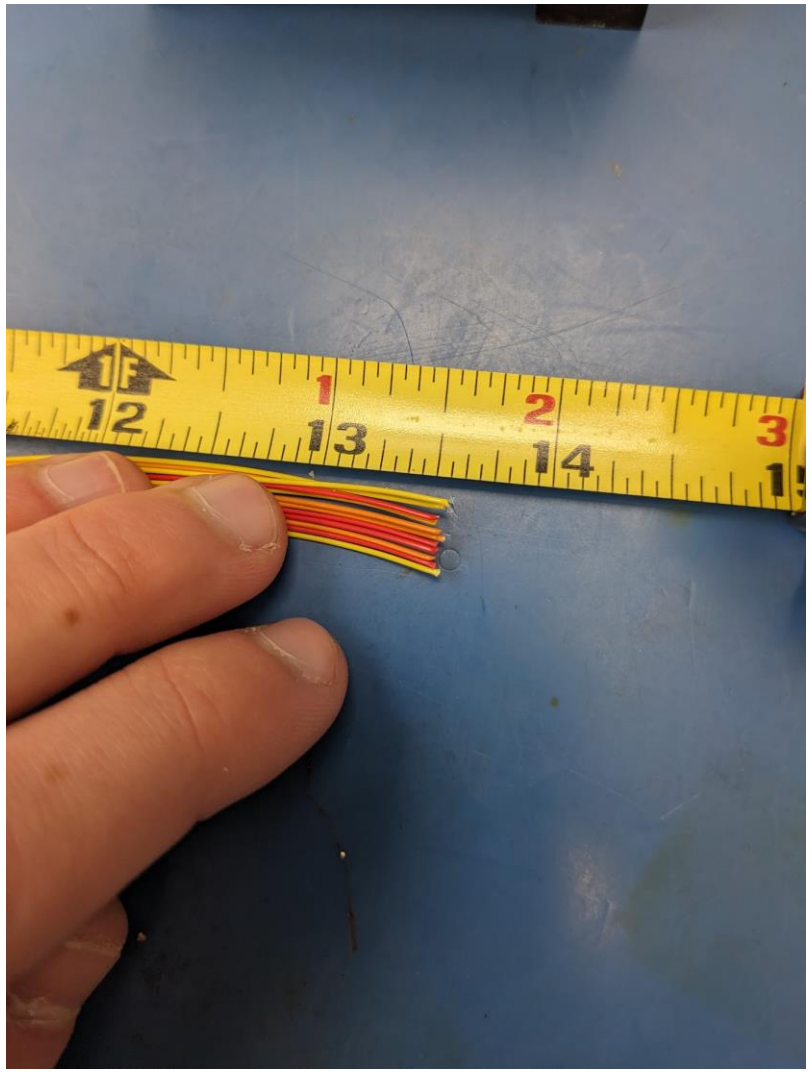
The cables should appear as shown below. The untrimmed length of red, orange, and yellow wires is >15”.

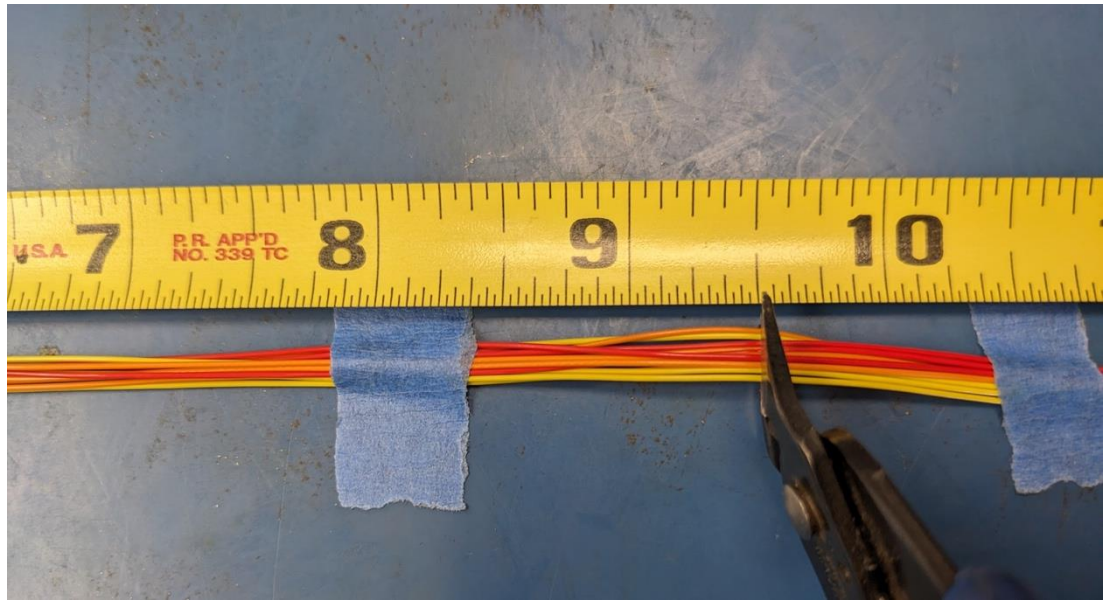
A completed wire harness consists of two cable bundles, one for the X-polarization and one for the Y-polarization. Each bundle needs to be trimmed to a different length.

The Y-polarization cable bundle should be 13.5”.

Measure from the end of the connector and cut so that 13.5” of wire remains.

Tolerances of up to 0.25” are acceptable.

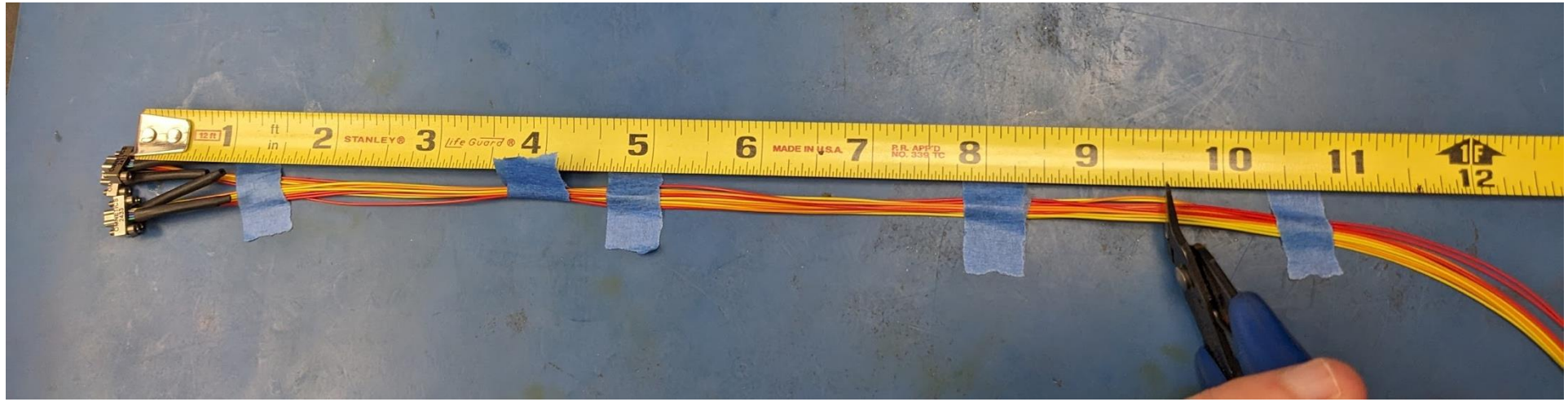




The X-polarization cable bundle should be 9.5”.

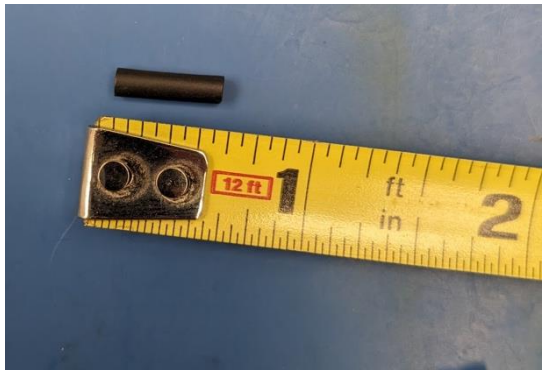
Measure from the end of the connector and cut so that 9.5” of wire remains.

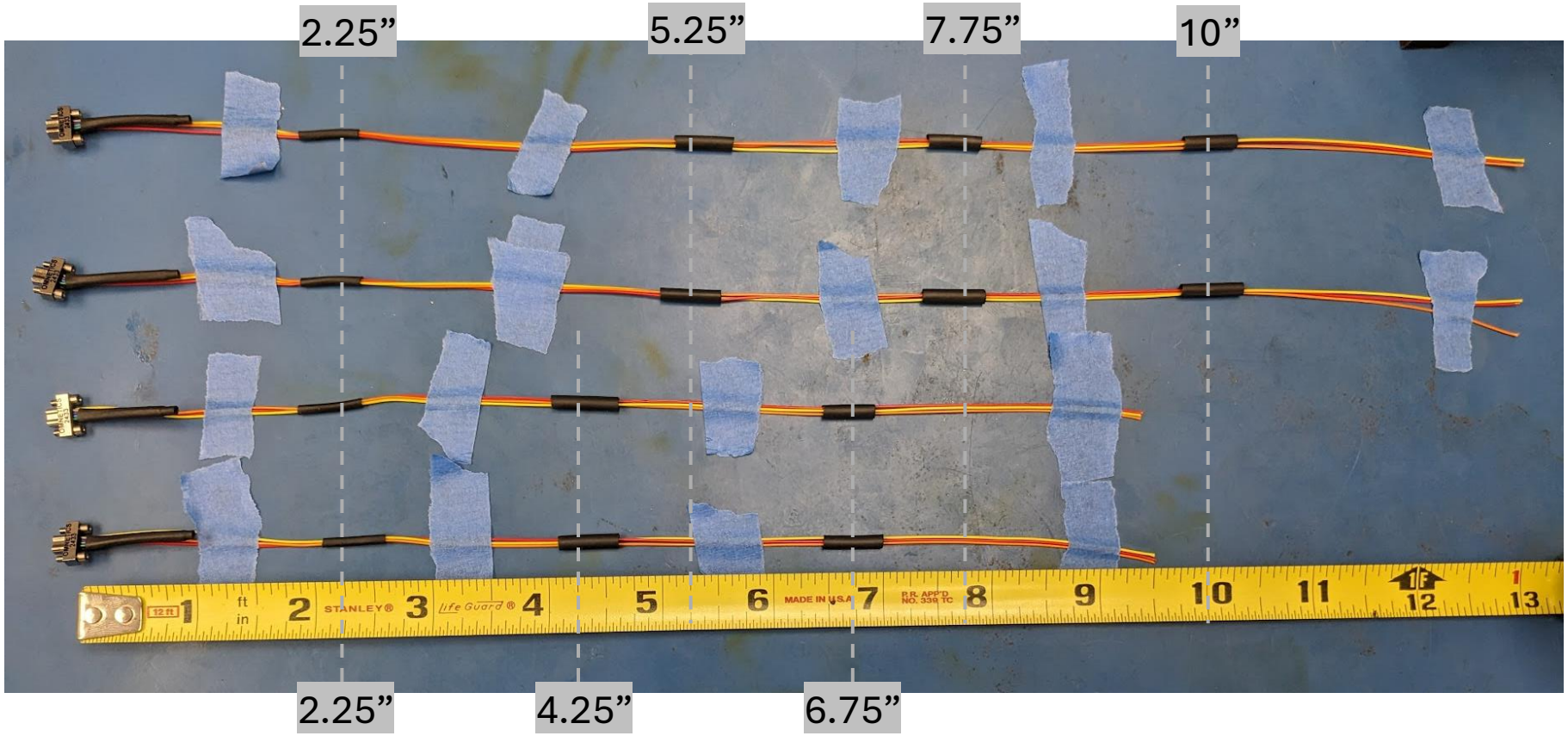
Tolerances of up to 0.25” are acceptable.



Cut 0.5”-long pieces of 3/32” shrink tube. Each X-polarization cable bundle (9.5”) will need 3 pieces, while each Y-polarization cable bundle (13.5”) will need 4 pieces.

Install the shrink tube pieces every 2-3 inches along the length of each cable wrap bundle as shown below. Exact placement is not too important. Approximate positions are annotated below. Apply the heat gun to seal.

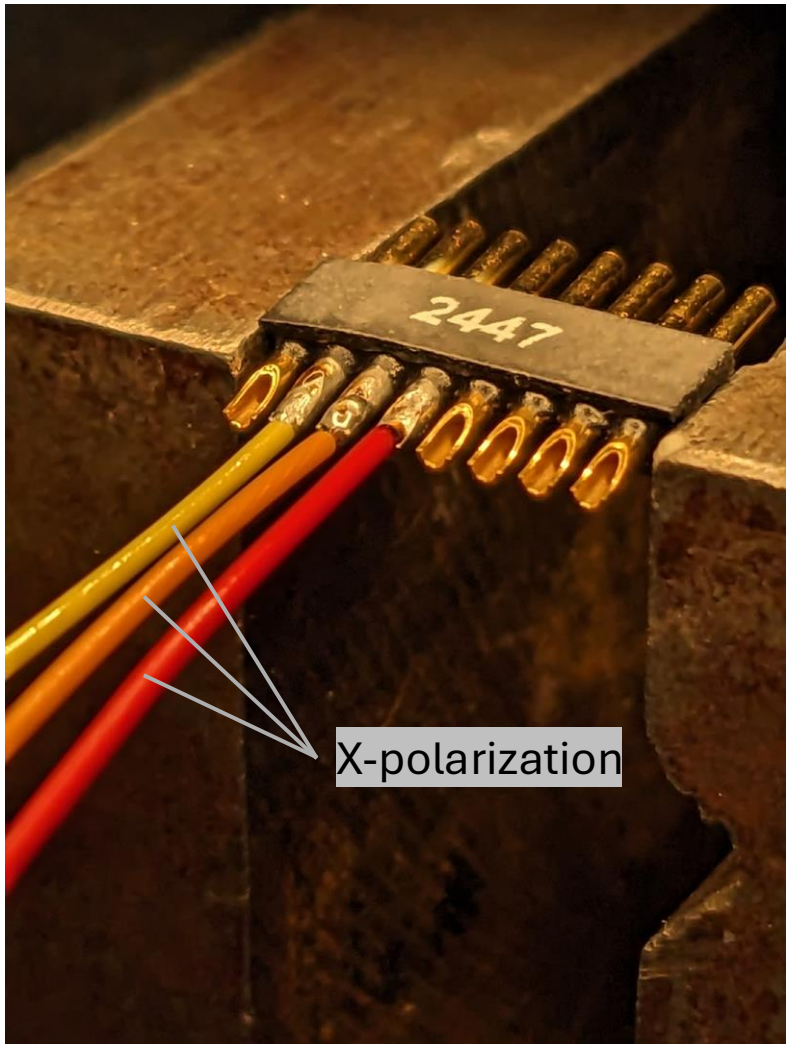


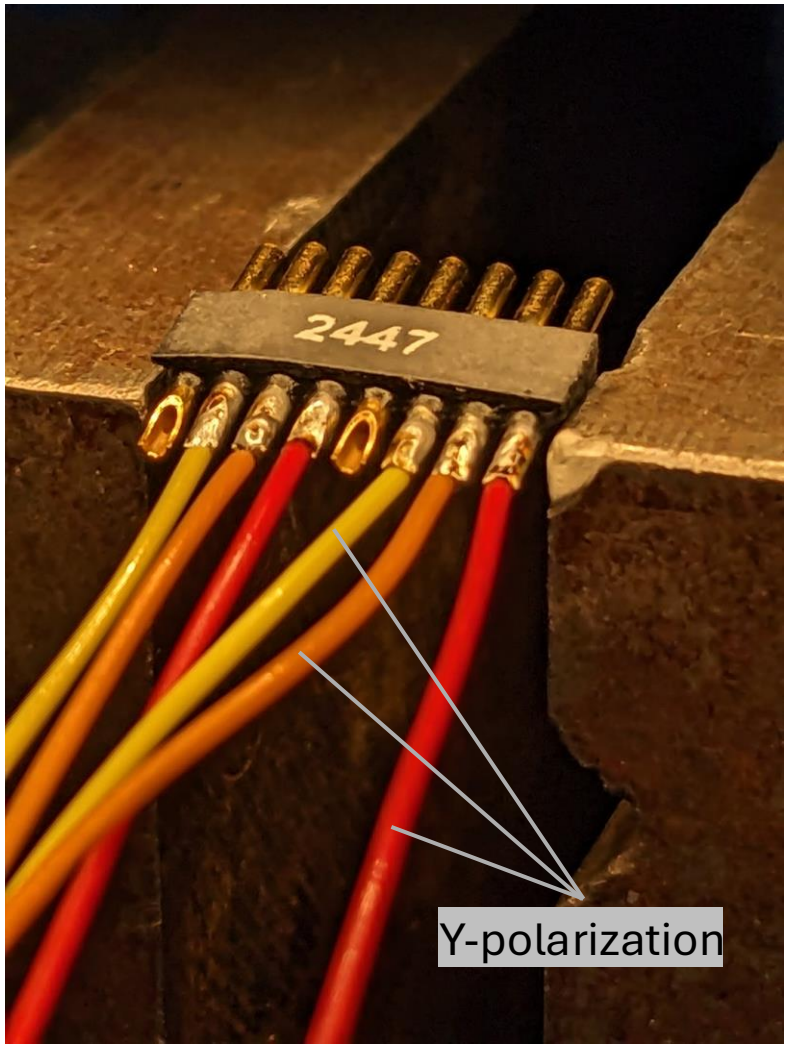
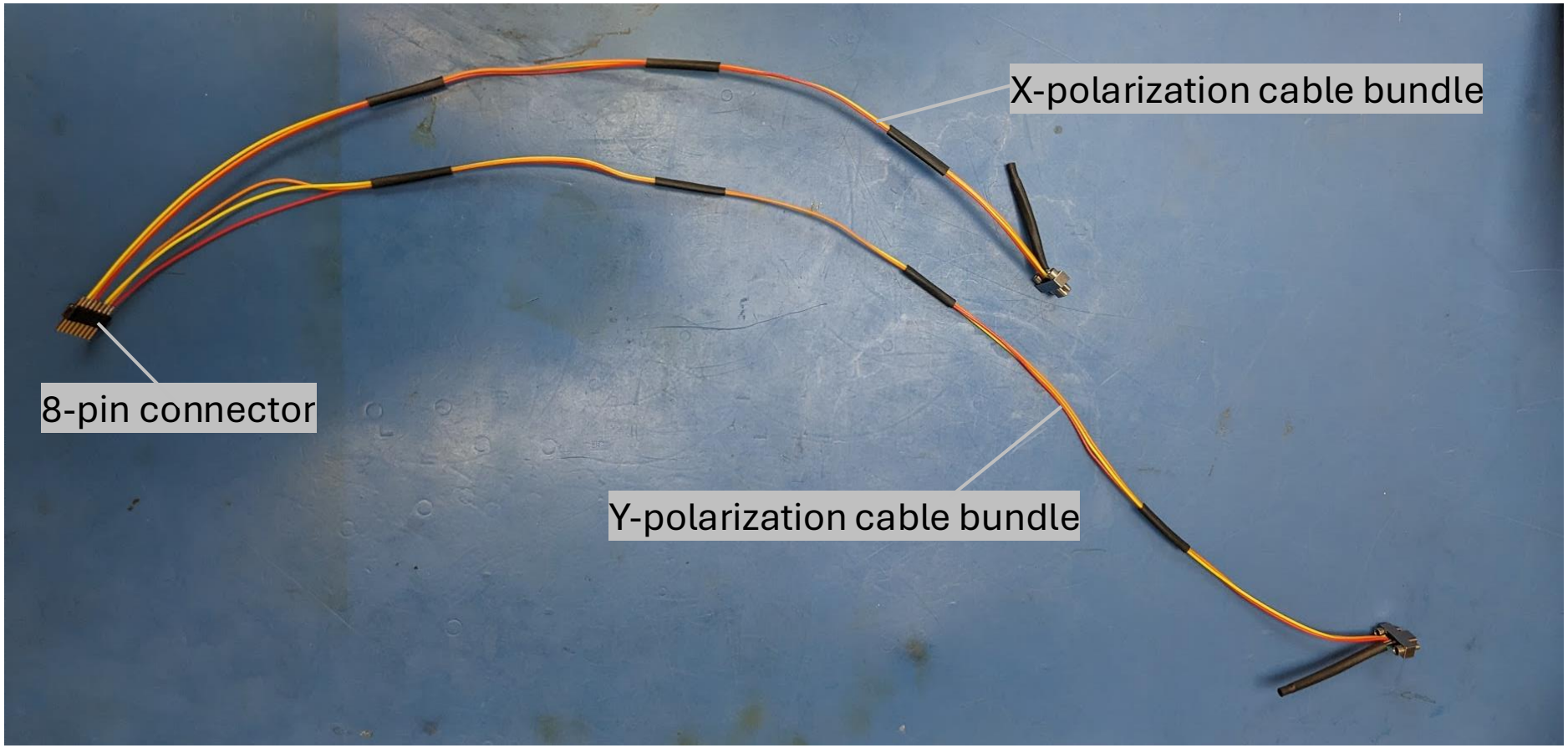


Strip a few mm of insulation from the end of each wire.

Insert the ends of the wires in the 8-pin connector in the order shown.
(empty pin, x-polarization yellow-orange-red wires, empty pin, y-polarization yellow-orange-red wires)

Solder in place.





A completed LNA wire harness should appear as shown below.